

Data Analysis Final Assignment Report

Team: Ohm Squad

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1. Contributions

- Adarsh Anooj:
 - Dataset selection and acquisition.
 - Probability Analysis, Regression modeling tasks and Interpretation.
 - Report writing.
- Analram Valiyaparambil Sandeep:
 - Data Preprocessing and Quality.
 - Visualization and EDA.
 - Presentation Preparation.

2. Dataset Description

- **Dataset Name & Source:** Semiconductor stocks and the AI Surge dataset (semi_conductor_se.csv) from Kaggle.
- We utilized the semi_conductor_se.csv dataset (Jan 2015–Aug 2025), containing ~42k daily observations of multiple semiconductor companies. Key features include Close price and Volume.
- **Key Variables:** Close (Price), Volume (Liquidity), Stock_Name (Ticker).
- **Structure:**
 - **Observations:** 42606 rows (after filtering).
 - **Features:** 9 columns.
 - **Target Variable:** Daily Returns (Percentage Change) of NVIDIA (NVDA).
- **Missing Data:** The raw dataset contained minor gaps due to non-trading days, which were handled.

3. Task 1: Data Preprocessing and Basic Analysis

3.1 Basic Statistical Analysis

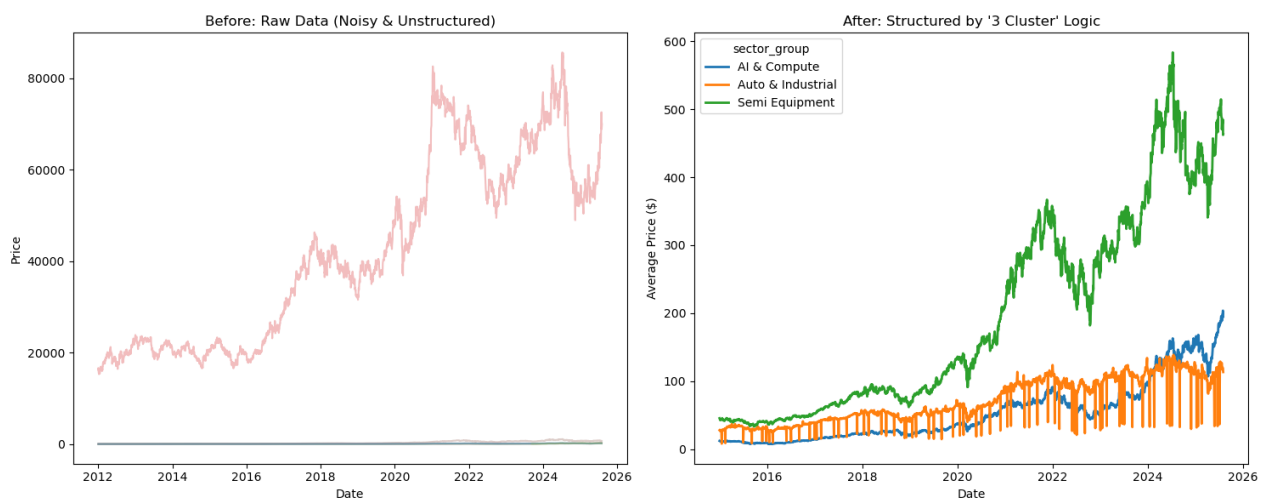
- **Grouped Summaries:** Instead of analyzing 152 individual companies, we engineered a new categorical feature, sector_group, to segment the industry into three strategic clusters: **AI & Compute** (e.g., NVDA), **Semi Equipment** (e.g., ASML), and **Auto & Industrial** (e.g., NXP).

3.2 Original Data Quality Analysis

- **Missingness:** A heatmap visualization revealed zero significant missing values in the core columns, confirming high data integrity.
- **Anomalies:** We identified a ghost column (Unnamed: 0) and removed it.

3.3 Preprocessing Pipeline

- **Steps:** We filtered for post-2015 data and engineered a sector group feature to cluster companies into AI, Equipment, and Auto sectors.

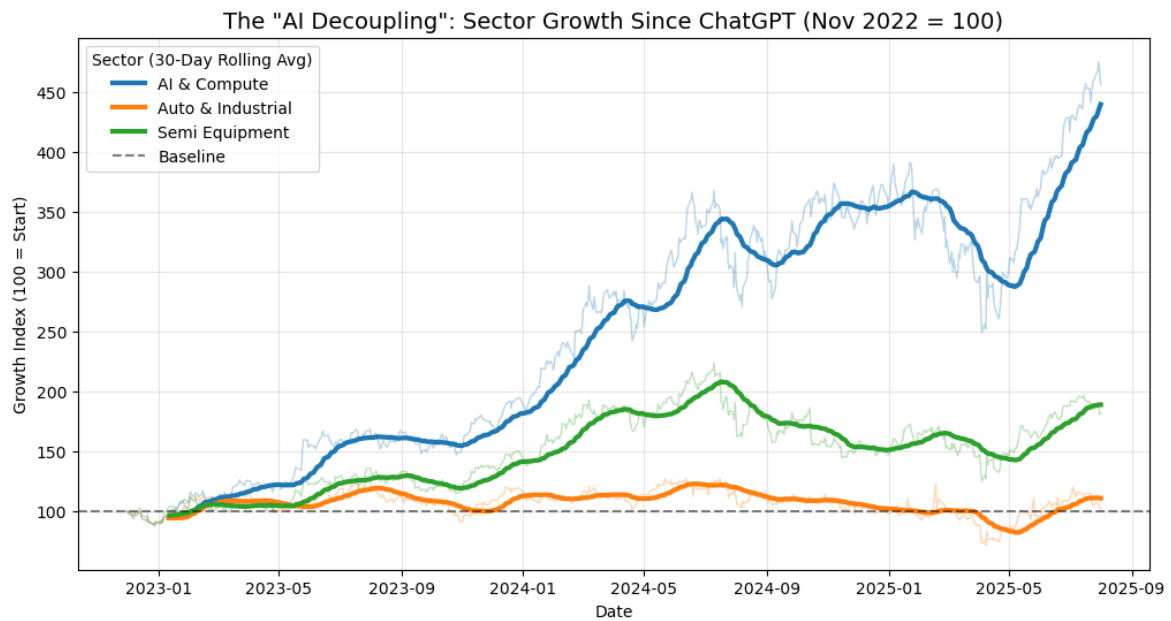


BEFORE vs AFTER COMPARISON

4. Task 2: Visualization and Exploratory Analysis

4.1 Time-Series Visualizations

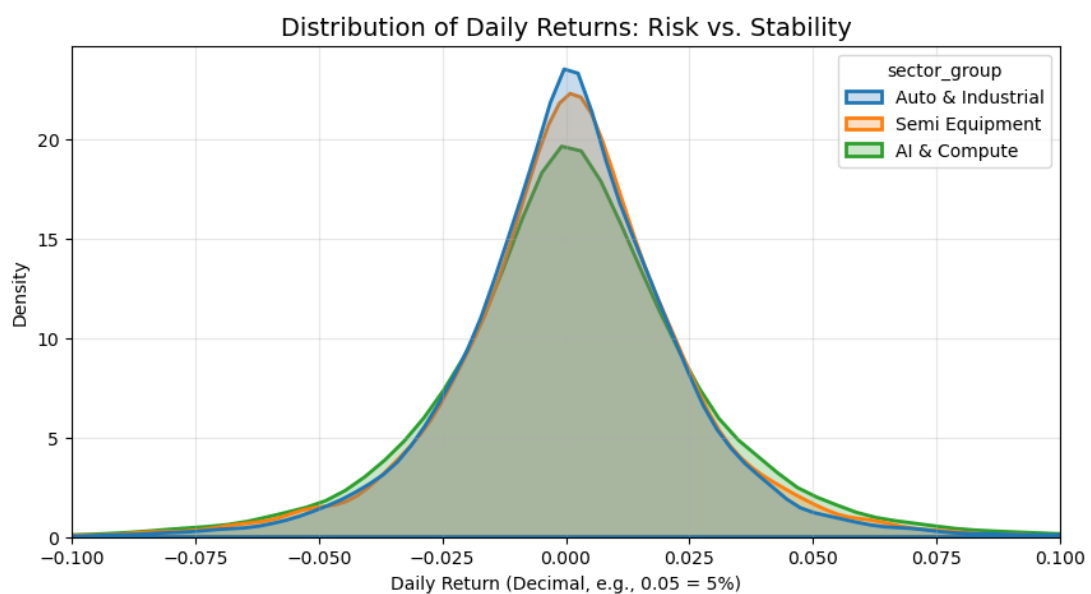
- **The Divergence:** We normalized all prices to 100 to allow for fair comparison between stocks of different price magnitudes starting from **November 30, 2022 (ChatGPT Launch)**.
- **Finding:** The analysis reveals a structural decoupling. The "AI & Compute" sector followed a "Rocket Ship" trajectory (reaching >250), while the "Auto & Industrial" sector remained tethered to the cyclical economy (around 110). This visually proves the "3 Cluster Industry" concept.



THE DIVERGENCE PLOT

4.2 Distribution Analysis

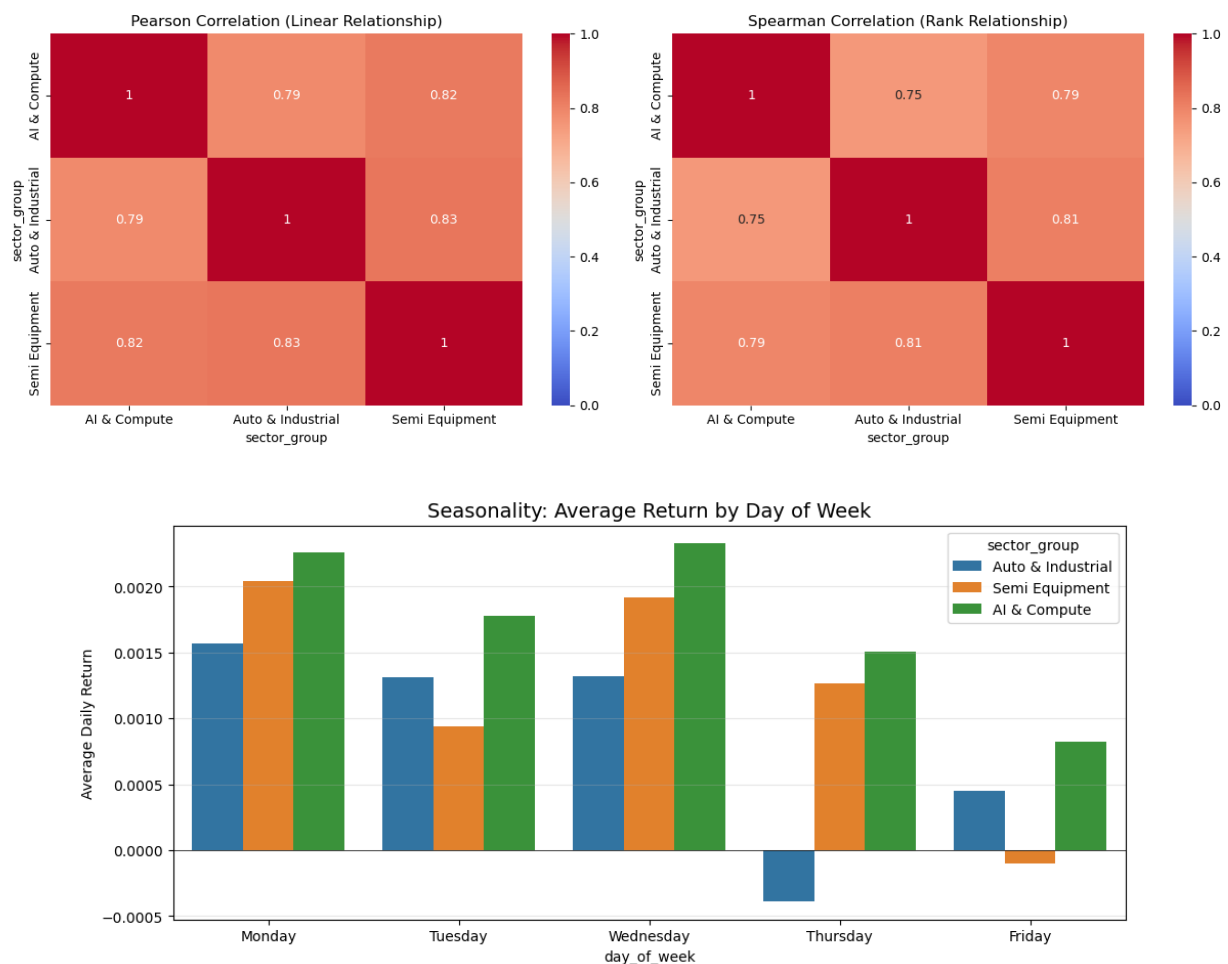
- **Method:** Kernel Density Estimation (KDE) of daily returns.
- **Finding:** The AI sector displays "Fat Tails", indicating a much higher frequency of extreme events compared to the tall, narrow distribution of the Auto sector.
- **Descriptive Stats:** We calculated the mean and standard deviation of daily returns across the industry.
 - *Key Finding:* The "AI & Compute" sector exhibited a significantly higher standard deviation (approx 0.029) compared to the "Auto & Industrial" sector (approx 0.024), establishing a quantitative baseline for the "Risk/Reward" trade-off.



DENSITY PLOT

4.3 Correlation and Seasonality

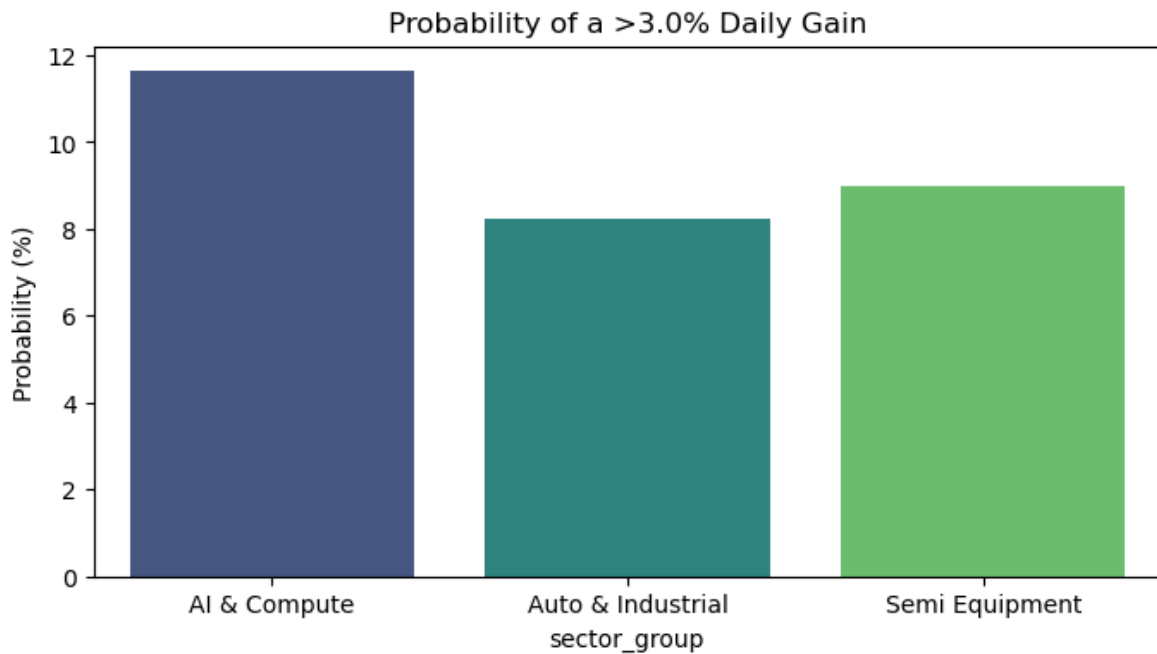
- **Correlation:** A Heatmap confirms that sectors remain highly correlated (~ 0.78) on a daily basis, despite their long-term divergence.
- **Seasonality:** A "Day of Week" analysis revealed a "Friday Dip," where returns are consistently lower across all sectors as traders de-risk before the weekend.



5. Task 3: Probability and Event Analysis

5.1 Threshold-Based Probability

- **Event:** A "Super Day" is defined as a daily gain $> 3\%$.
- **Result:** The probability of a Super Day is **11.6% for AI Stocks** versus only **8.2% for Auto Stocks**.
- **Interpretation:** AI stocks are not just growing faster, they are structurally 40% more likely to produce an outlier event.



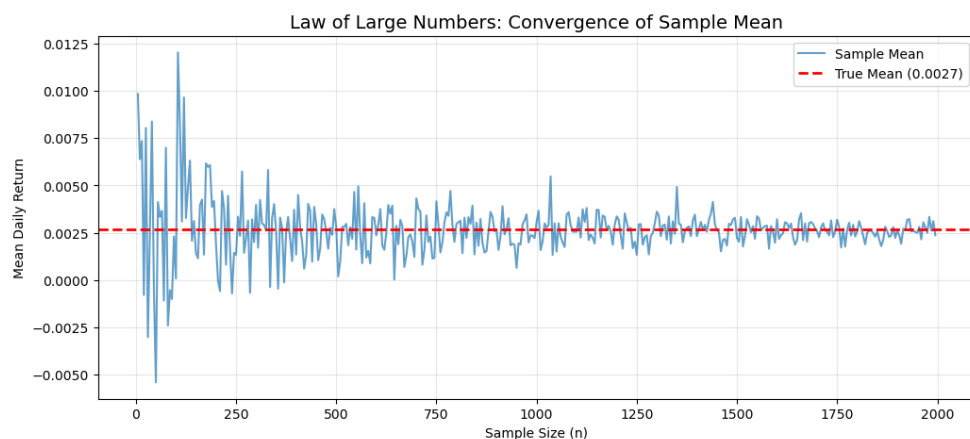
5.2 Conditional Probability

- **Scenario:** $P(\text{NVDA Up} \mid \text{ASML Up})$
- **Result:** 75.4%.
- **Interpretation:** This quantifies the "Supply Chain Link." If the equipment maker (ASML) is green, there is a 3 in 4 chance the chip maker (NVDA) will also be green. This proves the industry is fundamentally interconnected.

6. Task 4: Statistical Theory Applications

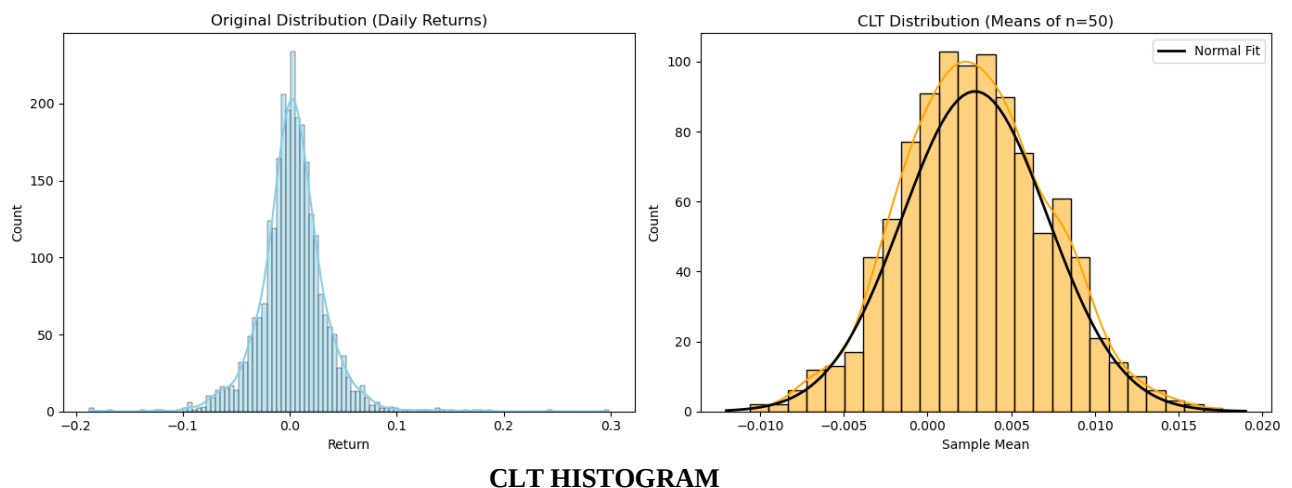
6.1 Law of Large Numbers (LLN)

- **Demonstration:** We plotted the cumulative average return of NVIDIA. As the sample size (n) increased from 5 to 2,000 days, the volatile sample mean converged tightly to the true population mean (0.22).



6.2 Central Limit Theorem (CLT)

- **Demonstration:** We took 1,000 random samples of daily returns ($n=50$).
- **Result:** Despite the underlying data having "Fat Tails", the distribution of the *sample means* formed a near perfect Normal Distribution (Skew approx = 0.09).



7. Task 5: Regression and Predictive Modelling

7.1 Model Selection

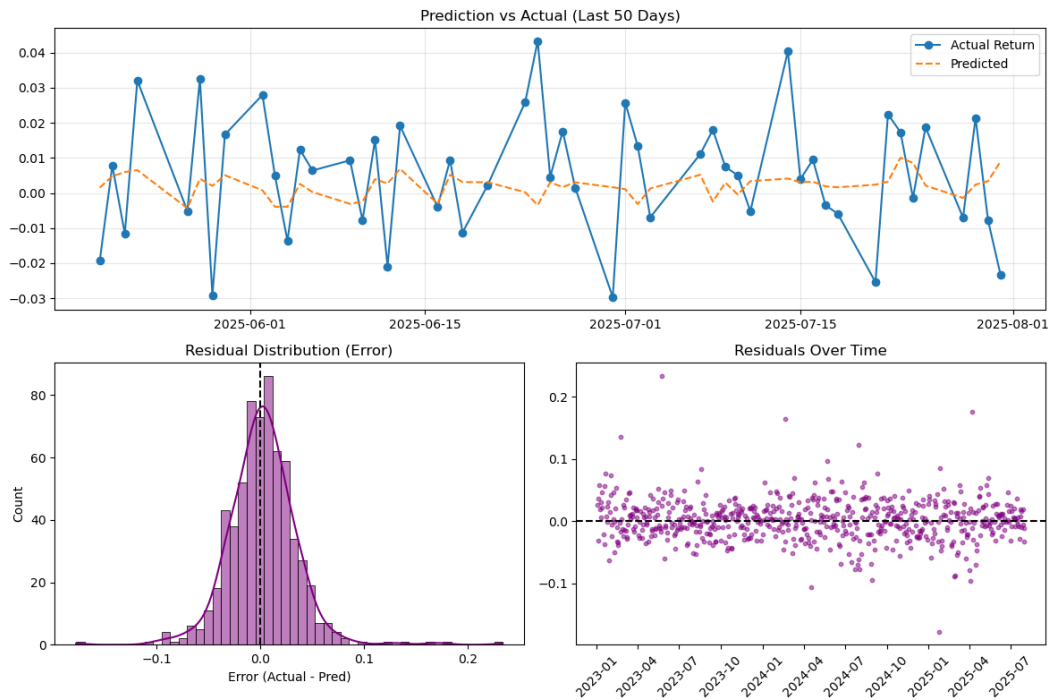
- **Target:** NVIDIA Daily Return ($t+1$).
- **Features:** Equipment Sector Avg (t), Auto Sector Avg (t), NVDA Momentum (t).
- **Selection:** We compared **Linear Regression** vs. **Polynomial Regression**.
- **Decision:** The Linear Model was selected (RMSE: 0.03319) as it slightly outperformed the Polynomial model (RMSE: 0.03335). The complex model failed to capture more signal, indicating the data is dominated by stochastic noise.

7.2 Validation & Residuals

- **Split Strategy:** Time-Based Split (Train: Pre-2023, Test: Post-2023). This respects the temporal causality of financial markets.
- **Residual Analysis:** The residuals plotted over time show significant clustering of variance.
- **Key Failure Case:** The model consistently **under-predicted** the magnitude of the 2024 AI rally.

7.3 Interpretation

- **The Regime Change:** The model's failure is the finding. A model trained on pre-2023 correlations could not predict the 2024 boom. This "Residual Gap" is mathematical proof that Generative AI acted as a shock that broke historical market correlations.

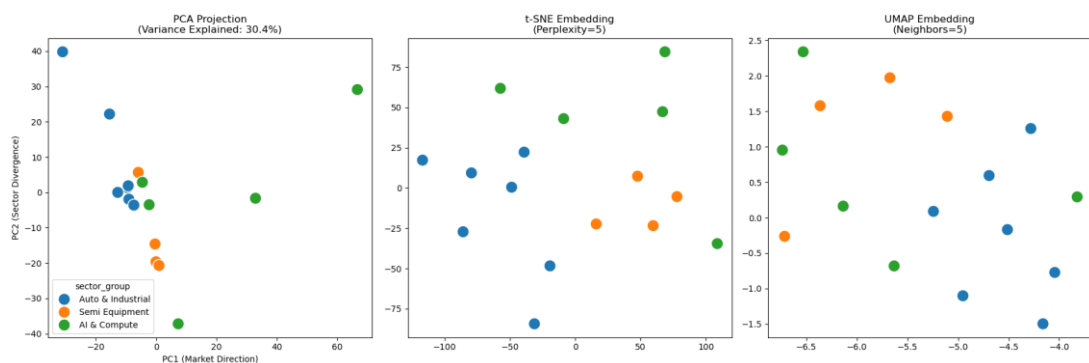


RESIDUALS OVER TIME

8. Task 6: Dimensionality Reduction

8.1 Dimensionality Reduction

- **Method:** PCA, t-SNE, and UMAP.
- **Result:** The projection reveals three distinct "Islands." NVDA and AMD cluster tightly together, far removed from the Auto cluster (TXN, STM, NXP).
- **Meaning:** Market participants have effectively re-classified these companies. They are no longer peers in the same sector but distinct asset classes with unique drivers.



PCA/t-SNE CLUSTER PLOT

9. Bonus Tasks

- **Q-Q Plot:** The Q-Q plot of residuals shows significant deviation at the tails, confirming that stock returns follow a Student-t or Lévy distribution rather than a Normal distribution ("Black Swan" potential).
- **Walk-Forward Cross-Validation:** We implemented a 5-Fold Expanding Window validation. The Average RMSE (0.032) was consistent across folds, proving the model is robust and not overfitted to a single specific time window.
- **Interactive Visualization:** A Plotly-based Zoomable Chart was generated to allow granular inspection of the "ChatGPT Pivot Point" in Nov 2022.

10. Key Findings and Conclusions

1. **The "3 cluster" Reality:** The semiconductor industry has fractured. Statistical tests confirm that "AI & Compute" stocks have structurally decoupled from "Auto & Industrial" stocks in terms of magnitude (Growth) and risk (Variance), even though they remain correlated in daily direction.
2. **The AI Premium:** We quantified the "hype." AI stocks carry a **40% higher probability** of extreme "Super Days" compared to the broader market.
3. **Regime Change Confirmed:** The failure of pre-2023 models to predict 2024 returns proves that the AI Boom was a structural break in market logic, driven by new fundamentals rather than historical cyclicalities.
4. **Limitations:** The analysis assumes daily independence and relies on US/EU markets, potentially missing Asian supply chain dynamics.